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Research Review on Human Activity Recognition Based on Wireless Sensing Technology

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Abstract. This review paper presents a comprehensive analysis of wireless sensing technology for non-contact human activity recognition (HAR), with a focus on applications in smart healthcare and public security. The study examines two primary sensing modalities: WiFi-based methods, including Channel State Information (CSI) and Received Signal Strength Indicator (RSSI), and radar-based approaches such as Frequency Modulated Continuous Wave (FMCW) and Ultra-Wideband (UWB). CSI-based systems leverage deep learning architectures like CNN-LSTM, DSC-SGRU, and Attention-GRU to achieve high precision and robust performance in complex environments, while RSSI-based techniques offer a cost-effective solution with lower hardware requirements, albeit with greater sensitivity to environmental variations. Despite their advantages, each method faces challenges, including limited generalization across heterogeneous scenarios, the trade-off between computational efficiency and accuracy, and difficulties in real-time validation. In addition, privacy and security concerns remain significant obstacles. To address these issues, future research should focus on multimodal data fusion, adaptive algorithm optimization for edge deployment, and the development of standardized, large-scale datasets. These efforts will pave the way for more reliable, efficient, and scalable HAR systems, ultimately enhancing the integration of wireless sensing in healthcare and security infrastructures. The insights provided in this review offer guidance for researchers and practitioners advancing HAR technologies.

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INTRODUCTION

In recent years, human activity recognition (HAR) has emerged as a crucial research domain driven by the rapid increase in global aging populations and the escalating demand for intelligent security systems [1]. Traditional methods relying on contact-based sensing with wearable devices often fail to meet the real-time and cost-effectiveness requirements of modern applications due to their high cost, low user comfort, and maintenance challenges. Meanwhile, computer vision-based techniques, despite their non-contact nature, are heavily affected by external factors such as lighting conditions and raise significant privacy concerns. Consequently, wireless sensing technology has attracted considerable attention as a promising non-invasive alternative for HAR, with applications spanning fall detection, gesture recognition, and broader smart healthcare and public security domains [2-3].

Wireless sensing systems exploit the inherent characteristics of wireless communication signals to extract critical information regarding human activities. In particular, two major categories of signals have been the focus of recent research: WiFi signals and radar signals. WiFi-based methods primarily involve Channel State Information (CSI) and Received Signal Strength Indicator (RSSI). CSI provides fine-grained physical-layer information that facilitates detailed analysis of human motion patterns, while RSSI offers a cost-effective approach due to its lower hardware complexity. On the other hand, radar-based approaches, such as Frequency Modulated Continuous Wave (FMCW) and Ultra-Wideband (UWB) technologies, have also been explored due to their ability to perform high-precision localization and tracking even in complex environments [4].

CSI-based systems benefit from advances in deep learning, with architectures such as CNN-LSTM, DSC-SGRU, and the lightweight Attention-GRU being developed to enhance recognition accuracy and adaptability in diverse settings. For instance, the CNN-LSTM model has demonstrated impressive performance with an average Euclidean error of 3.489% on datasets comprising various human actions, underscoring its potential in achieving high precision

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at a low deployment cost [5]. Nevertheless, despite these advantages, CSI-based methods still face challenges related to micro-motion recognition, computational overhead, and issues surrounding data standardization.

Similarly, RSSI-based techniques, which analyze signal strength variations through methods like Gaussian Kalman filtering integrated with the TAE-GRU model and the LCVAE-CNN model, offer the benefits of low hardware requirements and reduced computational complexity. However, they are typically more susceptible to dynamic environmental conditions and require substantial data annotation to maintain accuracy. These contrasting characteristics underline the importance of developing a comprehensive theoretical framework that can effectively compare and analyze the strengths and limitations of each modality.

This review systematically addresses the technical principles and methodological characteristics of wireless sensing systems. It aims to explore how hybrid deep learning architectures—incorporating feature fusion and computational optimization—can significantly improve recognition accuracy and environmental robustness. Moreover, the study identifies critical challenges, including limited generalization across heterogeneous scenarios, the inherent trade-off between computational efficiency and accuracy, and the difficulties of achieving real-time validation. In light of these challenges, future research should emphasize multimodal sensor fusion, the development of adaptive, edge-compatible algorithms, and the creation of standardized, large-scale datasets for more objective performance evaluation [2-3].

The following sections build on this framework to provide an in-depth comparative analysis of WiFi-based and radar-based HAR systems, thereby offering valuable insights for both researchers and practitioners striving to advance the field of non-contact human activity recognition.

COMPARATIVE ANALYSIS OF KEY TECHNOLOGIES

WiFi Signal-Based Sensing

WiFi sensing primarily utilizes two signal features: Channel State Information (CSI) and Received Signal Strength Indicator (RSSI) [4].

CSI extracts fine-grained physical-layer information from wireless channels to analyze human activities. Notable methodologies include the CNN-LSTM hybrid model, the DSC-SGRU hybrid mode, and the lightweight Attention-GRU model [5-7]. The CNN-LSTM model achieves an average Euclidean error of 3.489% in a dataset encompassing five actions (limb stretching, bending, hand-raising, right-hand lifting, and right-leg lifting), demonstrating improvements in both precision and low-cost deployment [5]. The DSC-SGRU model reduces parameters and computational costs by an order of magnitude compared to ResNet50 and ViT while maintaining 77.6% accuracy on the Widar dataset [6]. Despite advantages in precision, environmental adaptability, and multi-task compatibility, CSI signals face limitations in micro-motion recognition, computational overhead, and data standardization.

RSSI-based approaches analyze signal strength variations to recognize activities. Representative methods include Gaussian Kalman filtering integrated with the TAE-GRU model and the LCVAE-CNN model [8-9]. The TAE-GRU model outperforms conventional LSTM and GRU models in elliptical trajectory localization experiments (5,000 samples), demonstrating superior mean absolute error (MAE) and training efficiency [8]. The LCVAE-CNN model improves floor classification accuracy and mean positioning error (MPE) on the UJIIndoorLoc [10] and Tampere datasets compared to existing methods such as SAE-CNN and CDAE-CNN [9][11]. RSSI is suitable for low-cost deployments with moderate accuracy requirements but exhibits sensitivity to dynamic environments and demands high data annotation costs [12].

The advantages and application scenarios of WiFi signal are summarized in Table 1.

TABLE 1. The advantages and application scenarios of WiFi signal table.

Signal Type	Category	Typical Methods	Advantages	Application Scenarios
WiFi	CSI	DSC-SGRU CNN-LSTM Attention-GRU	High precision, strong environmental awareness, multi-task support	Complex environments requiring high accuracy and integrated perception-control systems
	RSSI	1.Gaussian Kalman Filter + TAE-GRU 2.LCVAE-CNN	Low hardware requirements and computational complexity	Scenarios with moderate accuracy demands, e.g., logistics and warehouse management

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FUTURE CHALLENGES

While wireless sensing technology has achieved remarkable advancements in non-contact human activity recognition, four critical technical barriers persist in its practical implementation:

1. Environmental Robustness Deficiency: Current algorithmic frameworks exhibit substantial performance degradation under complex operational conditions, particularly in multi-occupancy scenarios with overlapping signal patterns and dynamic acoustic interference. The stochastic nature of radio wave propagation in non-line-of-sight environments further exacerbates feature extraction challenges.

2. Computational Efficiency-Accuracy Tradeoff: High-precision models rely on GPU acceleration, limiting edge device deployment, while lightweight models sacrifice accuracy.

3. Data Diversity and Annotation Costs: Most datasets are lab-collected, resulting in high annotation costs and poor cross-scenario generalization.

4. Privacy-Security Trade-offs: Although less invasive than visual methods, wireless signals may still leak user location and behavior patterns, necessitating encryption and anonymization mechanisms.

RESEARCH PROSPECTS

Wireless sensing technology has already demonstrated considerable promise in non-contact human activity recognition, yet several challenges remain that hinder its full-scale deployment. Future research in this field should concentrate on addressing these obstacles by exploring multiple innovative directions, which are discussed below.

One major research direction is the integration of multimodal data. Currently, sensing methods based on WiFi signals (such as CSI and RSSI) and radar signals (including FMCW and UWB) each offer unique advantages. However, relying on a single type of signal often falls short in complex environments where occlusion, interference, and multipath effects can severely degrade performance. Future work should focus on combining multiple sensing modalities to achieve complementary strengths. For instance, merging the detailed environmental awareness provided by CSI with the high-precision positioning capabilities of UWB could lead to a more robust system capable of accurately tracking subtle human movements even in cluttered or dynamic settings. In addition, integrating inertial sensors could further enhance the recognition system by providing additional motion cues that are independent of the wireless channel variations. Such a multimodal fusion approach would not only improve overall recognition accuracy but also increase the system's resilience to environmental disturbances.

Another promising avenue is the development of adaptive algorithm optimization techniques. Presently, high-precision models typically require significant computational resources and often depend on GPU acceleration, which limits their applicability in edge devices or real-time systems. On the other hand, lightweight models, while computationally efficient, tend to compromise on accuracy. Future research should aim to develop adaptive algorithms that dynamically adjust network structures and hyperparameters in response to varying operating conditions. For example, incorporating dynamic network pruning, adaptive attention mechanisms, or layer-wise relevance propagation could allow the model to focus on the most informative features while reducing unnecessary computational overhead. Such adaptive strategies would help balance the trade-off between computational efficiency and recognition accuracy, enabling the deployment of high-performance models on resource-constrained platforms.

Standardized datasets and evaluation frameworks represent another critical area for future research. At present, many available datasets are collected in controlled laboratory settings, leading to issues with diversity and cross-scenario generalization. The development of large-scale, cloud-based datasets with unified formats that cover a wide range of real-world scenarios and populations is essential. Standardized datasets would facilitate a more objective comparison among different algorithms and promote the development of models that are more robust in heterogeneous environments. Moreover, leveraging semi-supervised or unsupervised learning techniques to reduce the cost and effort associated with data annotation could accelerate the development and deployment of these systems in practical applications.

Cross-disciplinary applications are expected to drive the next wave of innovation in wireless sensing. For instance, in the realm of virtual reality, wireless sensing could enable markerless motion capture, offering a more natural and immersive experience. In disaster rescue operations, radar-based sensing might be employed to detect and locate survivors hidden beneath debris, significantly enhancing emergency response capabilities. Similarly, applications in smart healthcare could benefit from non-invasive monitoring systems capable of detecting falls or abnormal behavior patterns in elderly individuals, thereby reducing response times and improving overall patient care. By fostering

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collaboration among experts in wireless communications, computer vision, machine learning, and domain-specific fields, future research can pave the way for integrated solutions that address complex, real-world challenges.

Hardware innovation is another crucial component that will shape the future of wireless sensing technology. Advances in integrated circuit design, sensor miniaturization, and low-cost material development are needed to reduce the overall expense and power consumption of sensing devices. The co-design of hardware and software—where system-level optimization is carried out to ensure efficient data acquisition, processing, and transmission—will be instrumental in developing compact and energy-efficient sensors. These improvements will enable the widespread deployment of wireless sensing systems in various settings, from smart homes and offices to public infrastructure and industrial environments.

The issue of privacy and security in wireless sensing systems should not be overlooked. Although wireless sensing is less invasive than visual surveillance methods, it still carries risks related to the inadvertent leakage of personal information such as location and behavioral patterns. Future research must prioritize the development of robust encryption and anonymization techniques to ensure that sensitive information is protected. Addressing these privacy concerns will be critical in gaining public trust and facilitating the broader acceptance of wireless sensing technologies in sensitive applications like healthcare and security.

CONCLUSION

As a pioneering methodology for non-contact human activity recognition, wireless sensing technology has garnered considerable attention across healthcare and public safety domains, primarily characterized by its cost-effectiveness, superior compatibility, and privacy-preserving nature. This paper conducts a systematic comparative analysis of perception paradigms leveraging WiFi signals (CSI/RSSI) versus radar-based systems (FMCW/UWB), elucidating their context-specific applicability. This paper highlights the synergistic advantages of hybrid deep learning architectures in simultaneously enhancing recognition accuracy, generalization capabilities, and environmental adaptability. Nevertheless, critical challenges persist, including limited generalization capacity under heterogeneous environmental conditions, inherent conflict between real-time processing demands and computational resource allocation, and dynamic operational constraints in practical deployments.

To bridge the gap between laboratory prototypes and industrial-scale implementations, subsequent research should prioritize: (1) multimodal sensor fusion strategies, (2) development of edge-compatible lightweight algorithms, (3) establishment of standardized benchmarking datasets, and (4) co-optimization of hardware-software ecosystems. These concerted efforts will not only facilitate the transition toward ubiquitous intelligent perception systems but also establish a comprehensive technical framework for sustainable elderly care solutions and next-generation urban security infrastructures through interdisciplinary collaboration.

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